

1. Design Database for Customer Purchase Application [DDL].

Description: The Customer Purchase Application (CPA) schema consists of several entities with specific purposes: **ProductCategory** stores product category details, **Product** stores product information linked to a category, **Address** stores shipping and customer addresses, **Customer** stores customer details linked to an address, and **PurchaseOrder** stores customer-specific product order information. The relationships between these entities include **One-to-Many** (e.g., one ProductCategory can have many Products, one Customer can have many PurchaseOrders), **One-to-One** (e.g., one Customer is linked to one Address), and potential **Many-to-Many** (e.g., customers ordering multiple products). Entities have constraints like NOT NULL for required fields, UNIQUE for unique values (e.g., ProductCode), CHECK for valid ranges (e.g., Price), and FOREIGN KEY to enforce relationships between tables.

Schema:

ProductCategory(CategoryId, CategoryName)

Product (ProductId, Name, Quantity, Discount, Price, ProductCategoryRef_Id, CreatedDate, ProductCode)

Column constraints: NOT NULL on Name, CHECK on Price, Foreign key on ProductCategoryRef_Id, UNIQUE on ProductCode)

Address (AddressId, City, Street, CountryName)

Customer (CustomerId, AddressRef_Id, Name, Address, EmailAddress, PhoneNumber)

PurchaseOrder (PurchaseOrderId, ProductRef_Id, CustomerRef_Id, OrderDate, PaymentTotal, OrderQuantity)

2. Write an SQL statement to perform the following modifications on existing tables **[DDL]**:

- i. Drop the CountryName column from the Address table.
- ii. Add a new column CountryRef_Id to the Address table, setting it as a foreign key that references the Country table.
- iii. Add a new column BillingAddressRef_Id to the PurchaseOrder table and set it as a foreign key that references the Address table.
- iv. Modify the default value of the Price column in the appropriate table.
- v. Remove an existing UNIQUE constraint on a column and then re-add it.

3. Write SQL queries to perform the following operations **[DML]**:

- i. Insert at least 5 records into each table that you created in Lab 1, using your own values.
- ii. Update one or more existing records in any table.
- iii. Delete one or more records from any table.

4. Rewrite the following SQL queries based on the Customer Purchase Application schema **[DQL]**:

- i. Write a query to retrieve the product codes of all products.
- ii. Write a nested query to retrieve the name and price of all products that belong to the 'TV' category.
- iii. Write a query using aggregate functions to find the maximum price, minimum price, and average price of all products.
- iv. Write a query to group products by their category and retrieve the categoryId, the number of products in each category, and their total price.
- v. Write a query using the HAVING clause to retrieve the categoryId, number of products, and their total price for each product category that contains more than two products.

5. Write the following queries in SQL, using the customer purchase application schema **[JOINS]**.
- i. Write a SQL query to get the name of each product along with the name of the category to which it belongs (Inner Join).
 - ii. Write a SQL query to retrieve the product name, product price, and product category name from the Product and ProductCategory tables, including all products even if they are not assigned to any category, and order the results by product name (Left Outer join).
 - iii. Write a SQL query to display the names of product categories that do not have any associated products (Right outer join).
 - iv. Write a SQL query to show all records from both the Product and ProductCategory tables. If there is no match found in either table, display NULL for the missing records (Full JOIN).
6. Create a database view that displays the total amount spent and the total number of orders for each customer.
7. Implement a transaction (with commit and rollback commands) in SQL.
8. Create a new user named “Ram” and Grant/Revoke privilege to insert records into product table.